

INTERNATIONAL news

World's first 3nm chips from Samsung

The 3-nanometer (nm) manufacturing node using the gate-all-around (GAA) transistor architecture by Samsung Electronics has begun. This production, when matched to its 5nm process, consumes 45% less power, offers improved performance by 23%, and has 16% smaller surface area. The company is also working on a second-generation 3nm process node that would be better in terms of power use, performance, and surface area.

Renesas and Cyberon to deliver integrated voice UI solutions

While Renesas makes a voice reference hardware platform available for rapid prototyping and development of voice interfaces, Cyberon's DSpotter solution eliminates the need for massive voice data collection while supporting over 40 global languages, empowering customers to adopt voice technology on embedded applications worldwide. Both companies have partnered to deliver voice user interface (VUI) solutions for Renesas' entire RA MCU line. It is a complimentary access being given to Renesas RA customers.

Bosch expanding facility to manufacture SiC semiconductors

Reutlingen is Bosch's silicon-on-chip (SiC) production facility. With increase in demand the company is expanding this facility between now and 2025. Bosch is looking to invest €3 billion by 2026, with around €400 million in the expansion of manufacturing capacity and the conversion of existing factory space. €250 million will be spent on adding 3000 square metres to the clean-room at the €1 billion Dresden fab it opened last year.

Delta Electronics sets up an R&D and manufacturing facility in Texas

Delta Electronics has acquired a nearly 30-acre complex in Plano, Texas, for R&D and manufacturing, that accommodates facilities of over 37,150sqm (400,000 square feet). The Plano complex is expected to facilitate Delta's expansion whilst creating hundreds of jobs. The new site is expected to operate 100% on renewable electricity by 2030 as the company hopes to potentially invest substantial resources in partnership with Texas universities and local colleges, to attract skilled talent in the region.

ST and GlobalFoundries to manufacture 300mm wafers

STMicroelectronics with GlobalFoundries Inc. have signed an MoU for a new jointly-operated 300mm semiconductor manufacturing facility in Crolles, France. The facility is targeted to ramp up to full capacity by 2026, with up to 620,000 300mm wafers per year whilst generating 1000 onsite additional staff. Significant financial support from France has been shown towards the companies for the new facility, which will strongly contribute to the objectives of the European Chips Act, including the goal of Europe reaching 20% of worldwide semiconductor production by 2030.

Europe's most powerful EV charging hub opened in Oxford

Oxford City Council, Pivot Power, Fastned, Tesla Superchargers and Wenea have officially launched Europe's most powerful EV charging hub that will offer fast and ultra-rapid charging for 42 vehicles at once at Oxford's Redbridge Park and Ride. This charging hub will be powered entirely by renewable energy. The project is part of a national network of Energy Superhubs developed by Pivot Power, which combines transmission-connected batteries and power infrastructure for EV charging to enable more renewables and accelerate the decarbonisation of transport.

Hyundai to set up first Korean EV factory

After 1996, Hyundai looks forward to opening its first factory in its origin country, South Korea, which will be solely dedicated to EVs. It is set to expand its sales by dedicated electric production. Hyundai has invested \$48 billion for this new plant apart from the EV production factory in Georgia, US. The production in Georgia will also include battery manufacturing and start in early 2023. The South Korean automobile manufacturing company is set to enter the electric vehicle market.

Panasonic to build second lithium-ion battery plant for EVs in US

The lithium-ion battery plant in Kansas could create up to 4000 new jobs as the Tesla supplier drives to expand its presence. The factory will be Panasonic's second electric car battery operation in the United States, joining its facility in Nevada. The new factory is expected to be located in De Soto, Kansas. The Kansas Department of Commerce said the state would reimburse Panasonic with estimated subsidies of \$829 million after the company has completed investment and hiring.

Volkswagen and Redwood collaborate to recycle EV batteries

Volkswagen and Redwood have joined hands for recycling EV batteries. Redwood would recycle EV batteries from Volkswagen and Audi in the United States by re-manufacturing battery electrode components from materials such as cobalt, nickel, lithium, and copper. It looks to work with more than 1000 Volkswagen and Audi dealers in the US to recover and recycle EV battery packs at its Nevada facilities. Volkswagen feels this partnership will help increase EV adoption such that 55% of its US vehicle sales will be fully electric by 2030.

Skyrora commences rocket production in the UK

By 2030, Skyrora plans to conduct sixteen launches per year from the Saxavord launch complex in the Shetland Islands alone. With the production of two Skyrora vehicles, which feature 3D-printed engine components that are already underway at the Cumbernauld site, the company plans to increase the total to sixteen per year once mass production begins. The company is also expected to generate new job opportunities, both in the North Lanarkshire area and in the rest of Scotland, with up to 100 highly-skilled technical and business roles.

It is in sync with the MeitY's vision of 1000 days. It was inaugurated by Union Minister of State, Electronics and Information Technology, Skill Development & Entrepreneurship, Rajeev Chandrasekhar, at Durgapur and Vardhman in West Bengal.

Chhattisgarh government introduces EV policy

The EV policy in Chhattisgarh exempts road tax on all EVs purchased during the first two years. Registration fee waivers are also being provisioned to encourage the

use of EVs, thereby ensuring relief to people bearing the brunt of rising fuel prices. A waiver in the registration fees on the sale of EVs along with the charging stations being built in government and private buildings will also be seen.